INCH-POUND

MIL-DTL-24308/33
w/AMENDMENT 1
20 November 2019
SUPERSEDING
MIL-DTL-24308/33
1 April 2015

DETAIL SPECIFICATION SHEET

CONNECTORS, ELECTRIC, RECTANGULAR, MINIATURE, COMBINATION INSERT ARRANGEMENT, PLUG, PIN CONTACTS, SHELL SIZE 3, SIZE 20 SIGNAL CONTACTS AND SIZE 8 POWER AND SHIELDED, CLASS G, M, N AND D

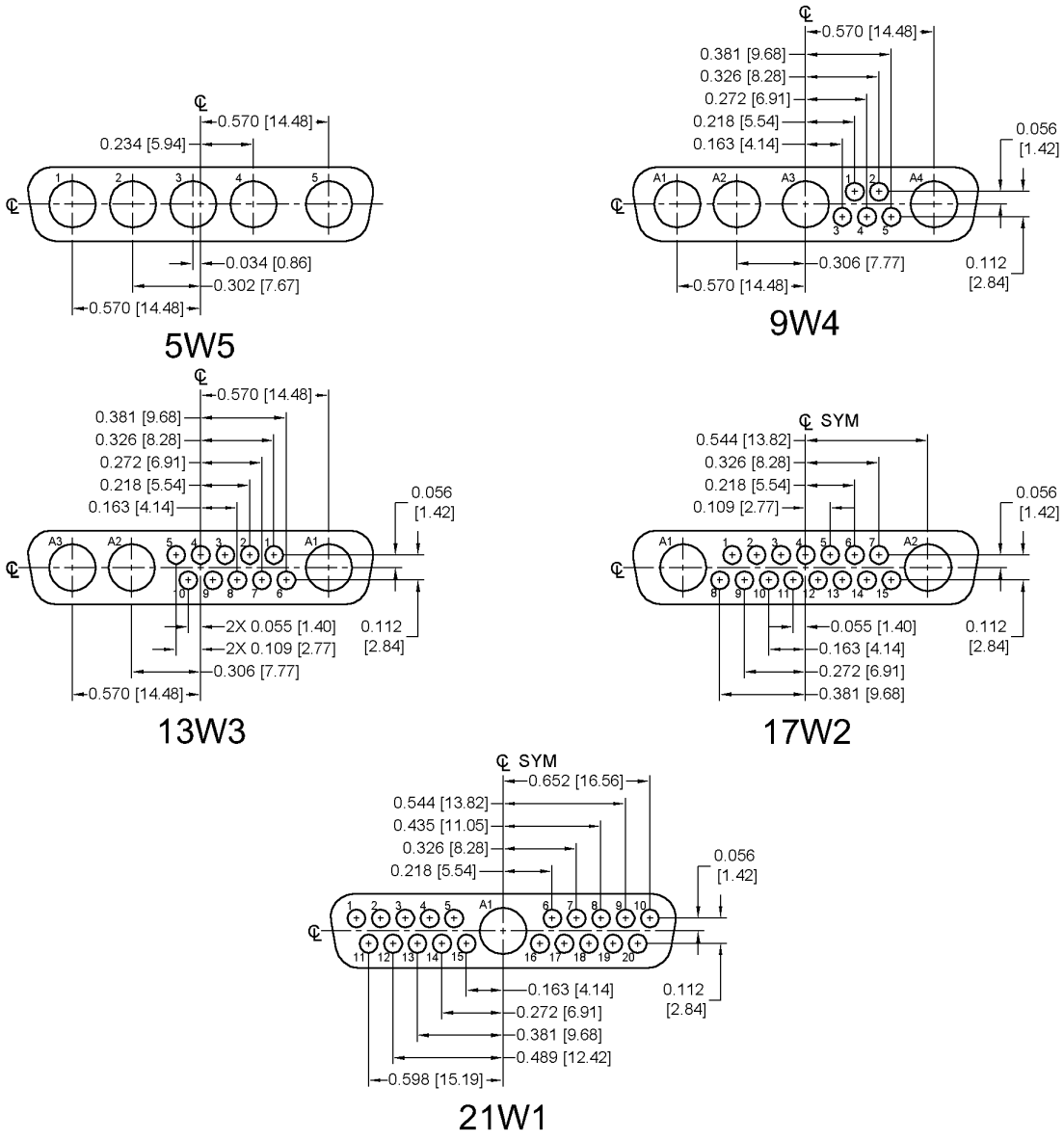
This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24308.

AMSC N/A

FSC 5935

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NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).
4. All dimensions are true position (TP).
5. Contact holes in insert shall be located on TP within a .005 inch (0.13 mm) diameter circle.
6. Reference documents shall be of the issue in effect on date of invitations for bid.
7. 2WK2 have 1 pin and 1 socket contacts. Socket contact shall be loaded in A1 position.

FIGURE 1. Insert arrangements, shell size 3, pin contacts.

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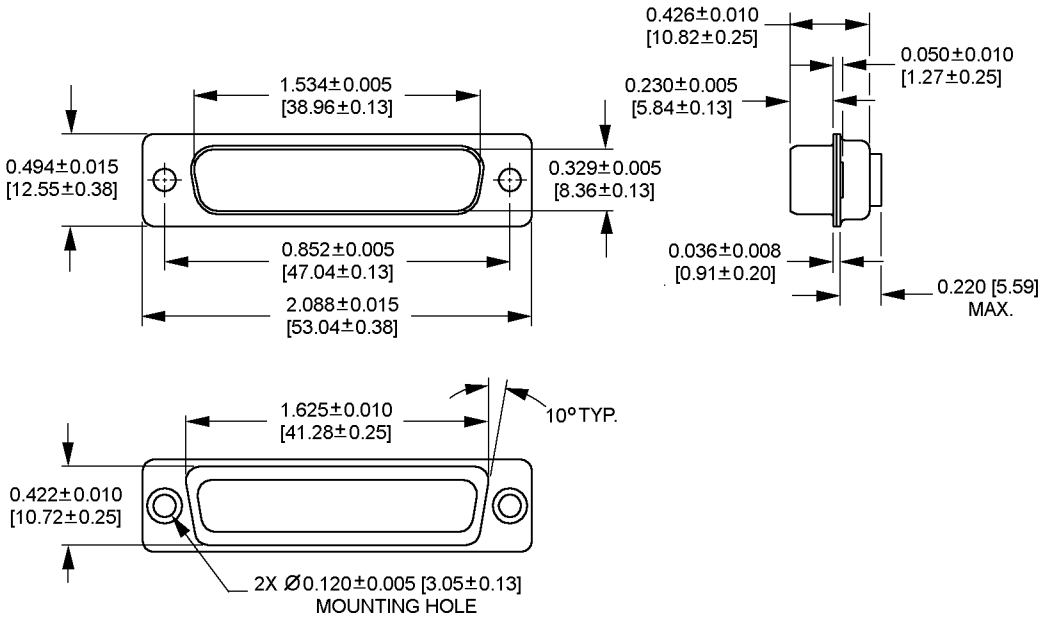
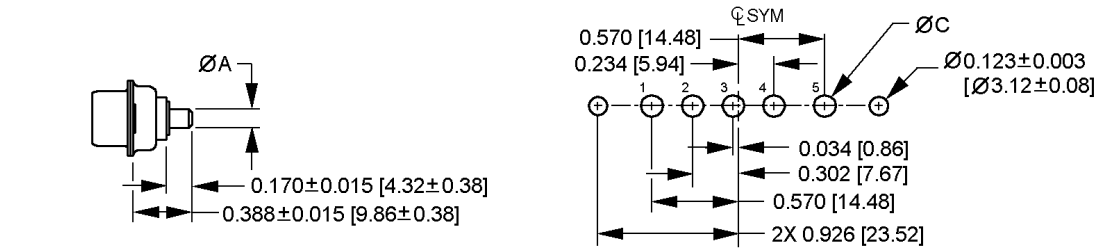
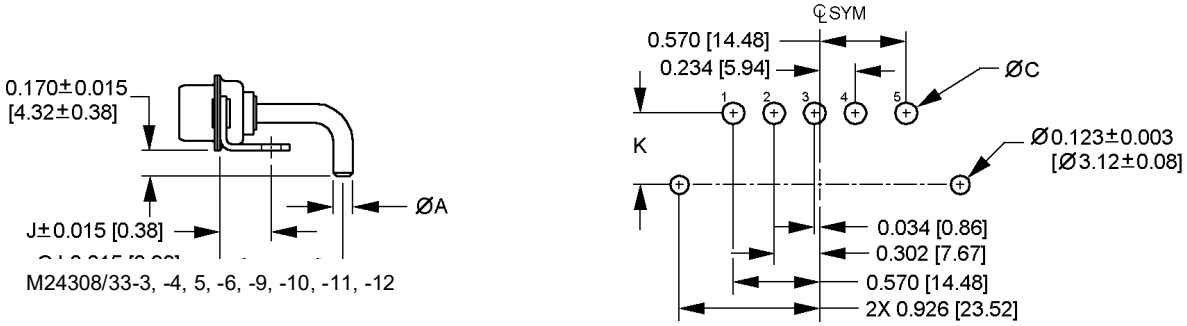


FIGURE 2. Dimensions shell size 3.

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M24308/33-1, 2, -7, -8

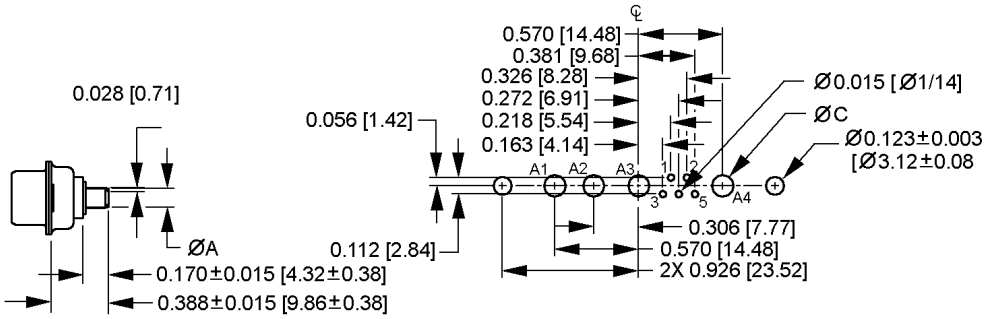


M24308/33-3, -4, 5, -6, -9, -10, -11, -12

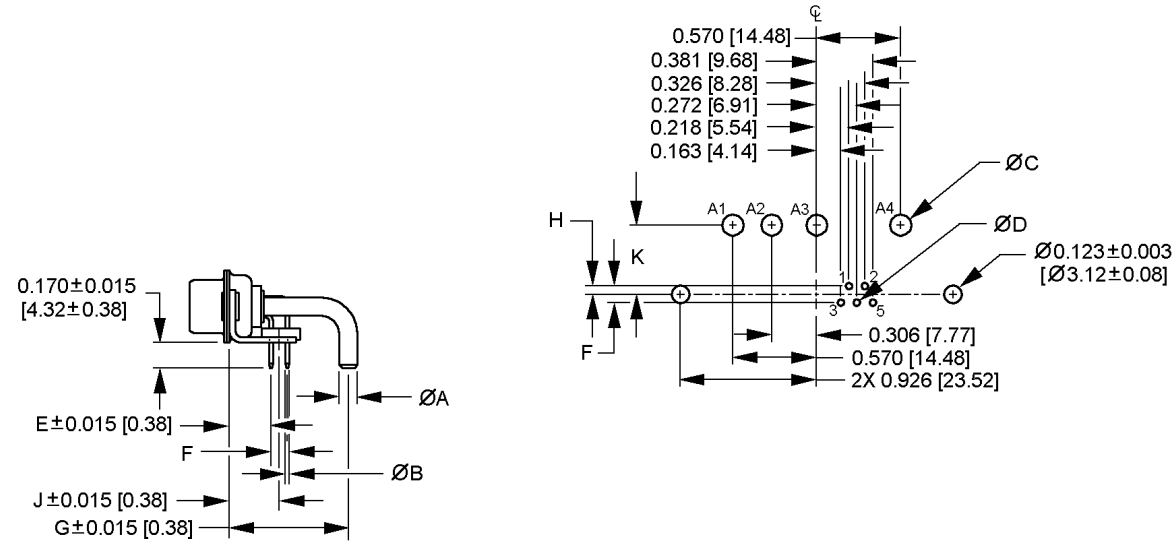
5W5

FIGURE 3. Termination dimensions and layout arrangement for, power connector.

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M24308/33-101, -102, -107, -108

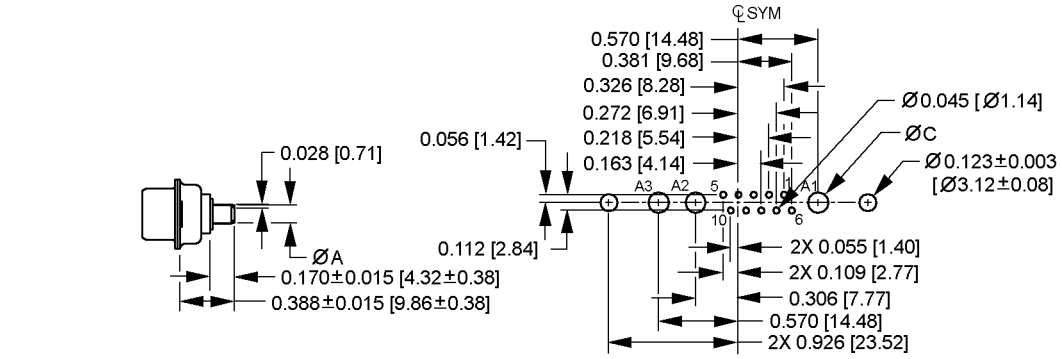


M24308/33-103, -104, -105, -106, -109, -110, -111, -112

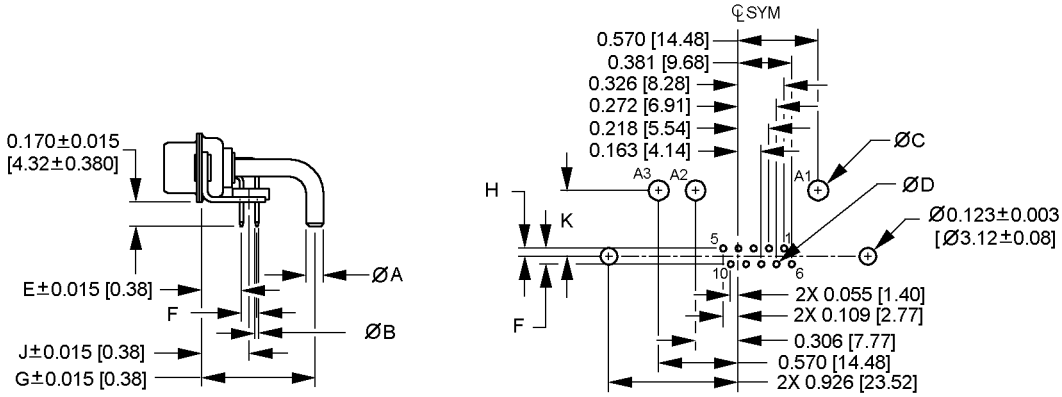
9W4

FIGURE 3. Termination dimensions and layout arrangement for, power connector - Continued.

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M24308/33-201, -202, -207, -208

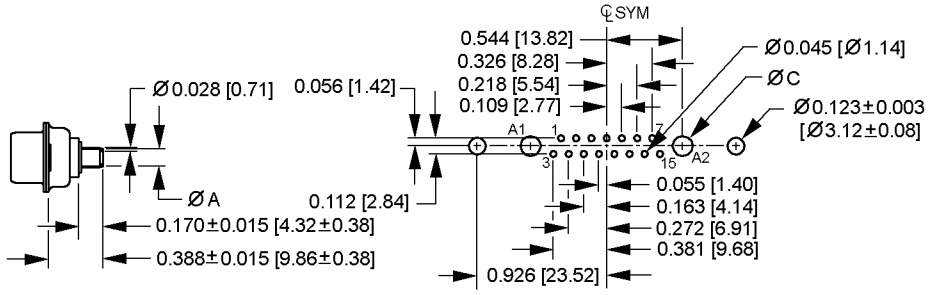


M24308/33-203, -204, -205, -206, -209, -210, -211, -212

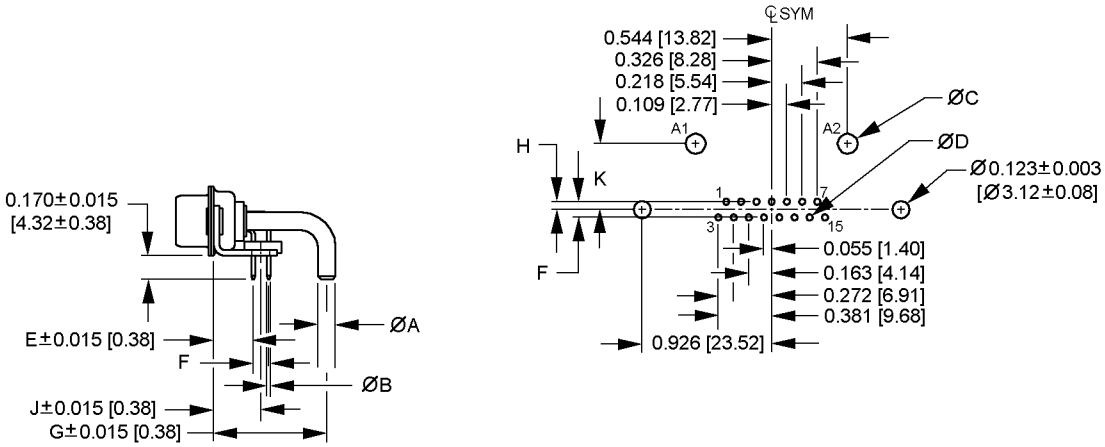
13W3

FIGURE 3. Termination dimensions and layout arrangement for, power connector - Continued.

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M24308/33-301, -302, -307, -308

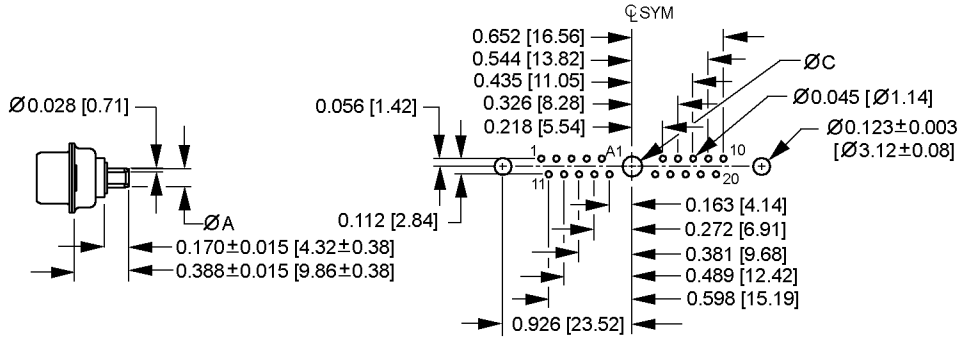


M24308/33-303, -304, -305, -306, -309, -310, -311, -312

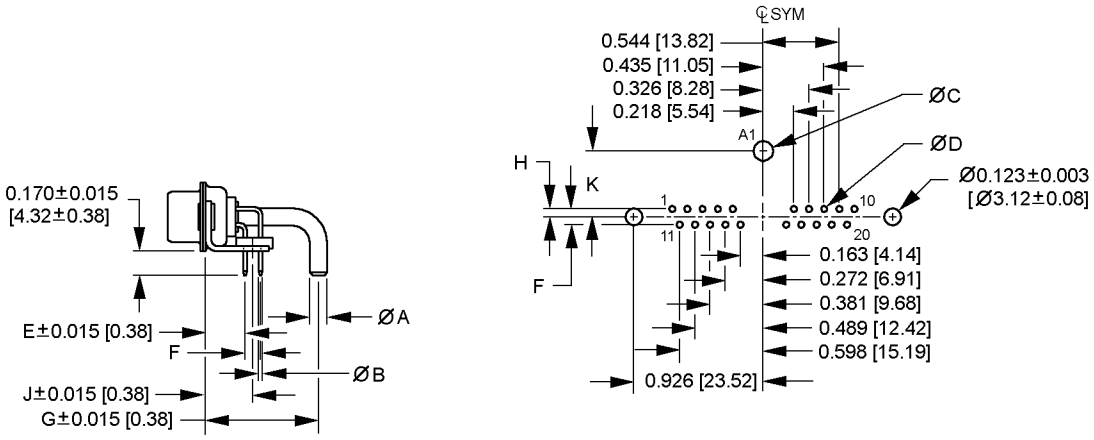
17W2

FIGURE 3. Termination dimensions and layout arrangement for, power connector - Continued.

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M24308/33-401, -402, -407, -408



M24308/33-403, -404, -405, -406, -409, -410, -411, -412

21W1

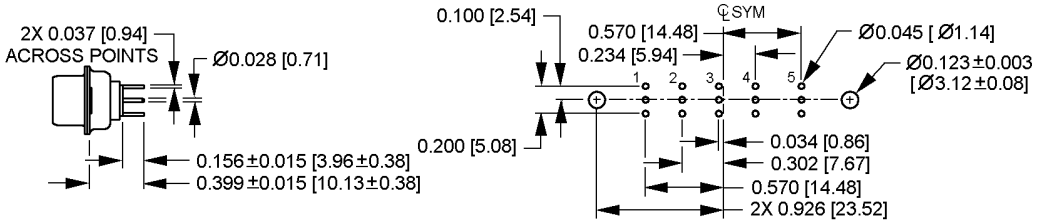
FIGURE 3. Termination dimensions and layout arrangement for, power connector - Continued.

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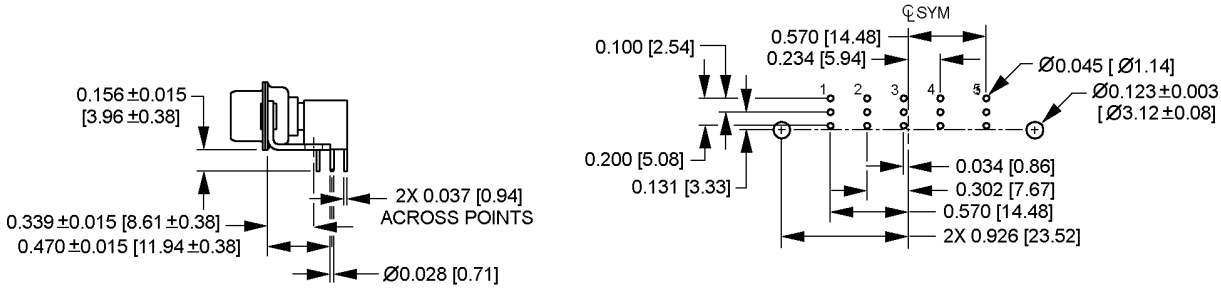
PIN M24308/33	ØA	ØB	ØC	D	E	F	G	H	J	K
-001, -002 -101, -102	0.078 (1.98)	---	0.098 (2.49)	---	---	---	---	---	---	---
-007, -008 -107, -108	0.125 (3.18)	---	0.145 (3.68)	---	---	---	---	---	---	---
-003, -004 -113, -114	0.078 (1.98)	---	0.098 (2.49)	---	---	---	0.339 (8.61)	---	0.339 (8.61)	0.000 (0.00)
-009, -010 -109, -110	0.125 (3.18)	---	0.145 (3.68)	---	---	---	0.810 (20.57)	---	0.339 (8.61)	0.471 (11.96)
-005, -006 -105, -106	0.078 (1.98)	---	0.098 (2.49)	---	---	---	0.420 (10.67)	---	0.420 (10.67)	0.000 (0.00)
-011, -012 -111, -112	0.125 (3.18)	---	0.145 (3.68)	---	---	---	0.810 (20.57)	---	0.420 (10.67)	0.390 (9.91)
-201, -202 -301, -302 -401, -402	0.078 (1.98)	0.028 (0.71)	0.098 (2.49)	0.045 (1.14)	---	---	---	---	---	---
-207, -208 -307, -308 -407, -408	0.125 (3.18)	0.028 (0.71)	0.145 (3.68)	0.045 (1.14)	---	---	---	---	---	---
-203, -204 -303, -304 -403, -404	0.078 (1.98)	0.028 (0.71)	0.098 (2.49)	0.045 (1.14)	0.283 (7.19)	0.112 (2.84)	0.339 (8.61)	0.056 (1.42)	0.339 (8.61)	0.000 (0.00)
-209, -110 -309, -310 -409, -410	0.125 (3.18)	0.028 (0.71)	0.145 (3.68)	0.045 (1.14)	0.283 (7.19)	0.112 (2.84)	0.810 (20.57)	0.056 (1.42)	0.339 (8.61)	0.471 (11.96)
-205, -206 -305, -306 -405, -406	0.078 (1.98)	0.024 (0.61)	0.098 (2.49)	0.040 (1.02)	0.370 (9.40)	0.100 (2.54)	0.420 (10.67)	0.050 (1.27)	0.420 (10.67)	0.000 (0.00)
-211, -212 -311, -312 -411, -412	0.125 (3.18)	0.024 (0.61)	0.145 (3.68)	0.040 (1.02)	0.370 (9.40)	0.100 (2.54)	0.810 (20.57)	0.050 (1.27)	0.420 (10.67)	0.390 (9.91)

FIGURE 3. Termination dimensions and layout arrangement for, power connector - Continued.

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M24308/33-015, -016

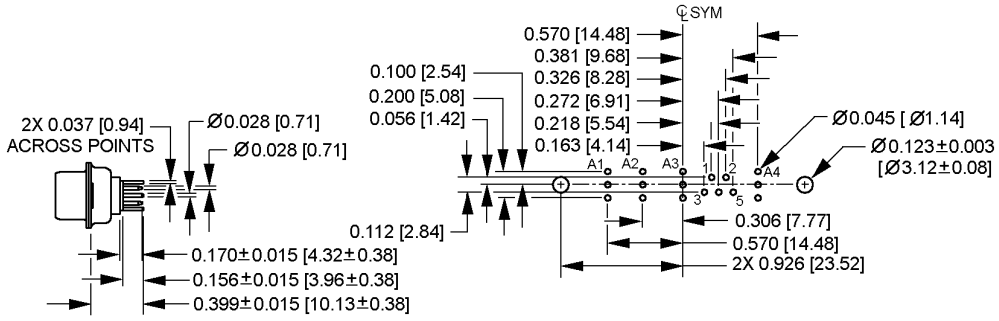


M24308/33-013, -014

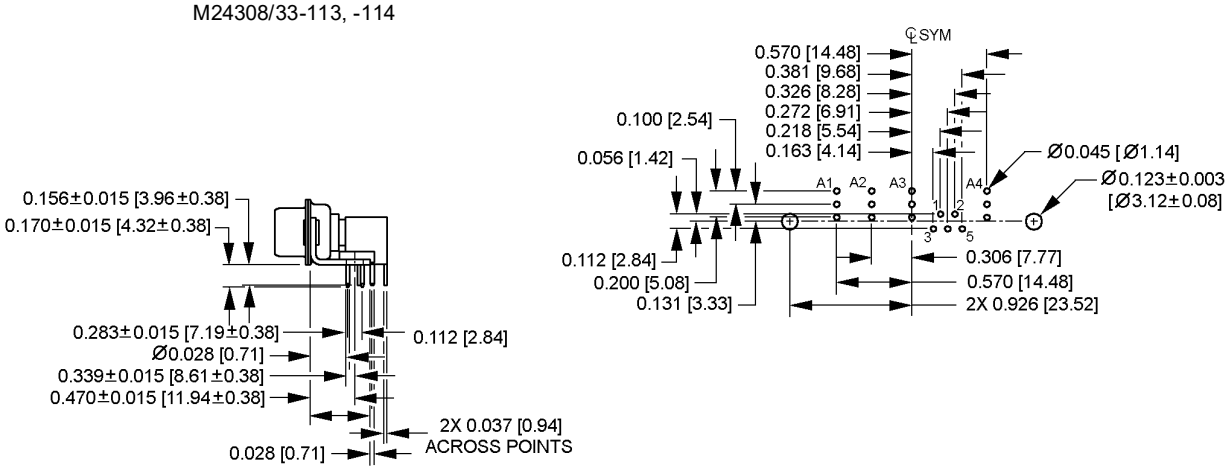
5W5

FIGURE 4. Termination dimensions and layout arrangement for, shielded connector.

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M24308/33-113, -114

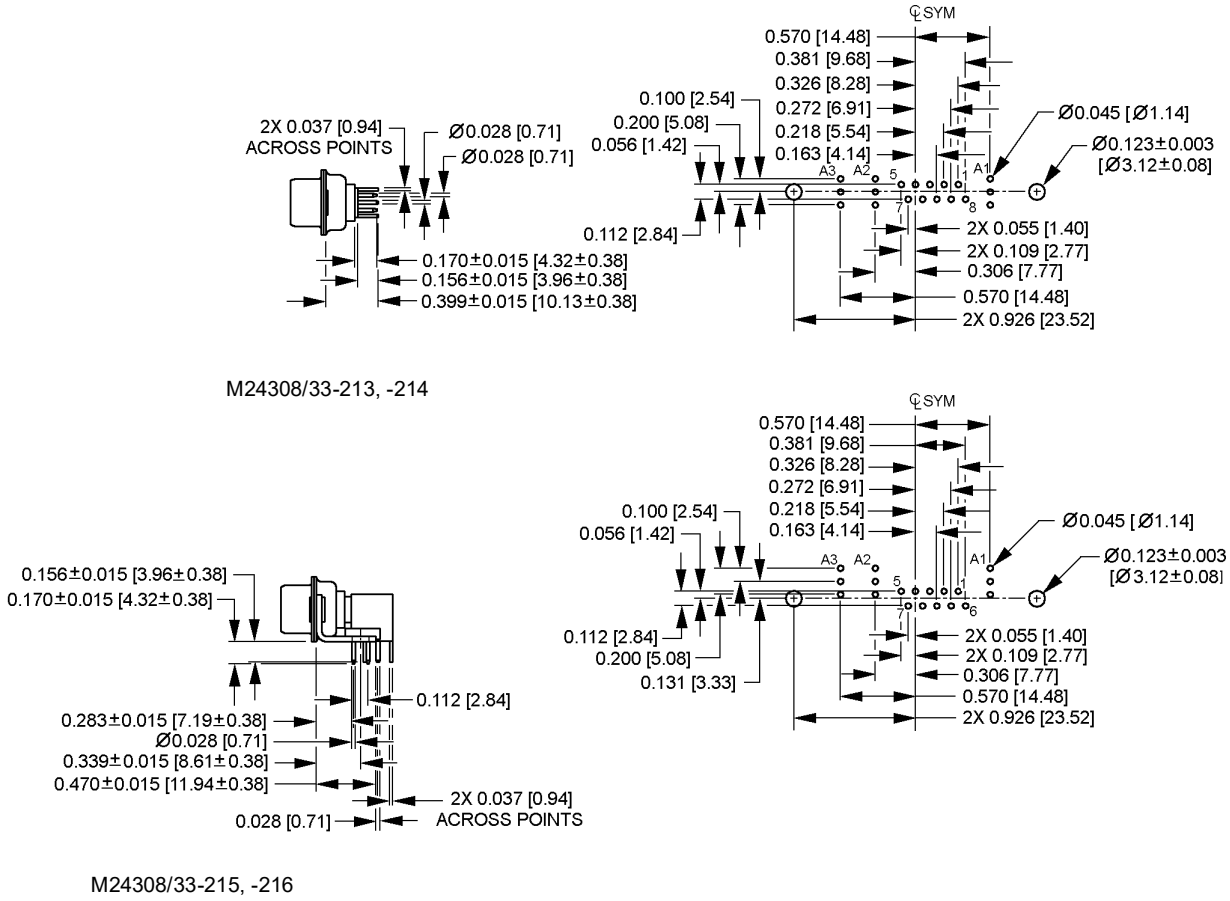


M24308/33-115, -116

9W4

FIGURE 4. Termination dimensions and layout arrangement for, shielded connector - Continued.

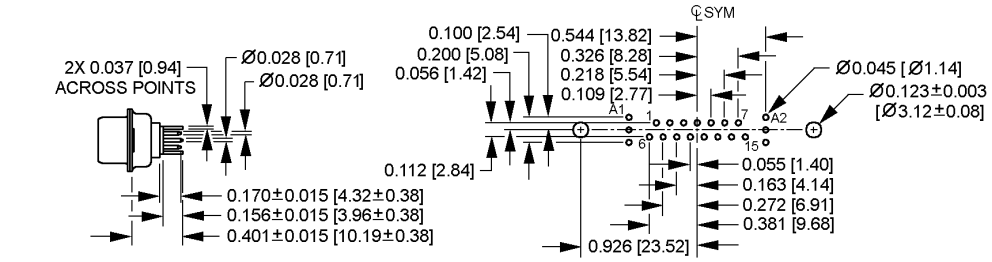
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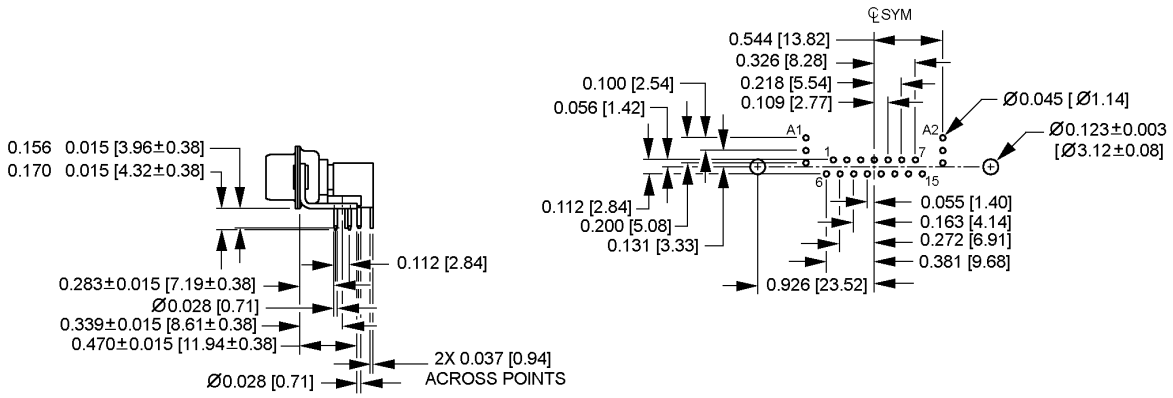
13W3

FIGURE 4. Termination dimensions and layout arrangement for, shielded connector - Continued.

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M24308/33-313, -314

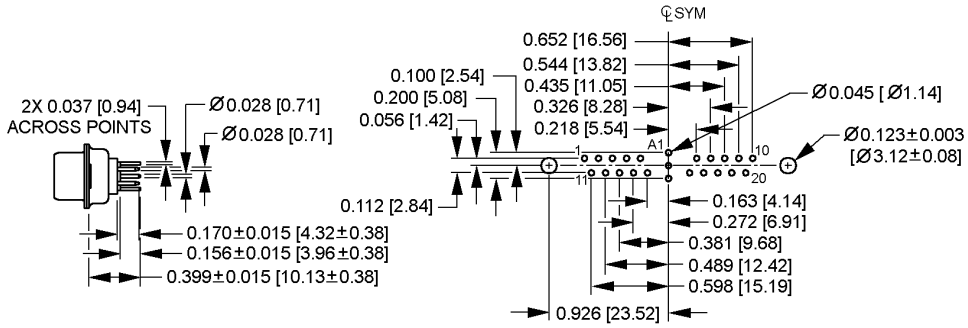


M24308/33-315, -316

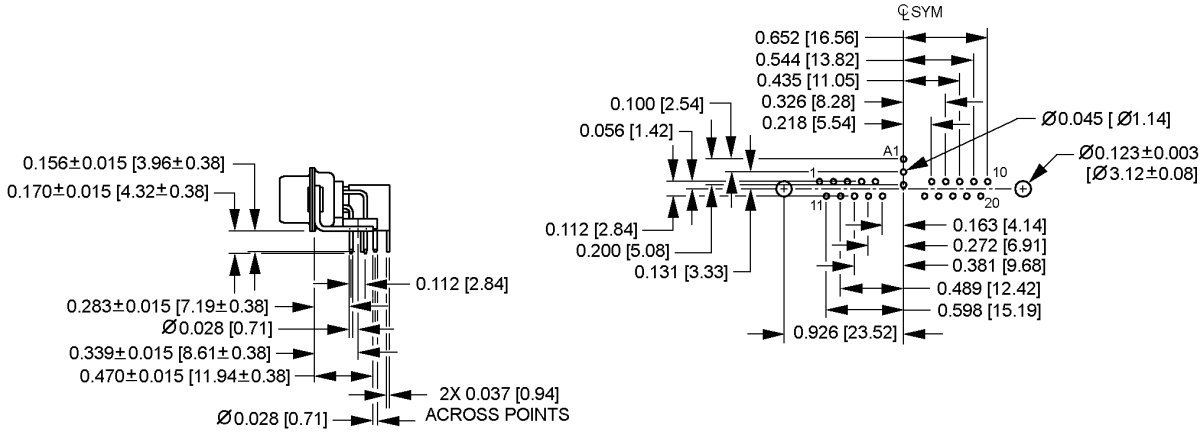
17W2

FIGURE 4. Termination dimensions and layout arrangement for, shielded connector - Continued.

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M24308/33-413, -414



M24308/33-415, -416

21W1

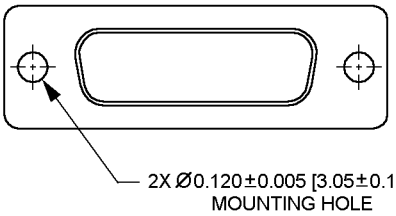
NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).

FIGURE 4. Termination dimensions and layout arrangement for, shielded connector - Continued.

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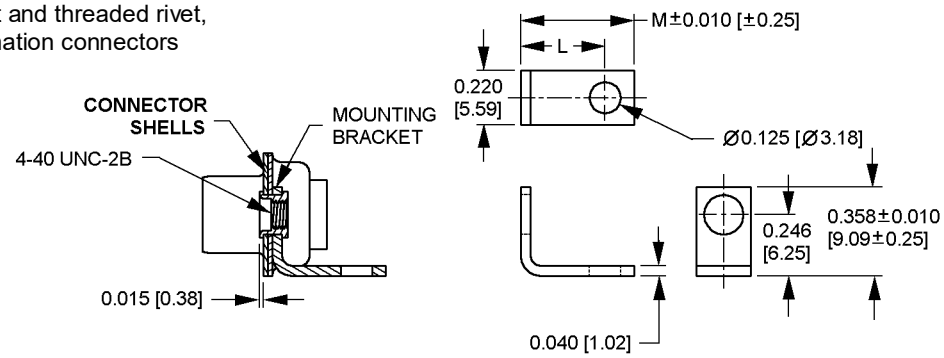
Configuration A:

Ø .120 (3.05 mm) mounting hole, straight termination connectors



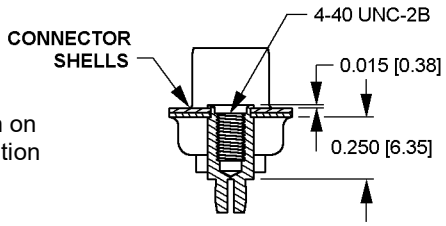
Configuration B and C:

Mounting bracket and threaded rivet, right angle termination connectors



Configuration D:

Swaged spacer with push on fastener, straight termination connectors



Configuration E and F:

Mounting bracket and threaded rivet with push on fastener, right angle termination connectors

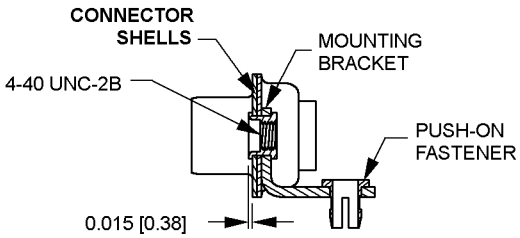


FIGURE 5. Mounting configurations.

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Mounting bracket configuration	L	M	Mounting type
B	0.339	0.456	0.125 hole
C	0.420	0.537	0.125 hole
E	0.339	0.456	Fastener
F	0.420	0.537	Fastener

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Unless otherwise specified, tolerance for three place decimals is ±.005 inch (0.13 mm).

FIGURE 5. Mounting configurations - Continued.

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REQUIREMENTS:

Dimensions and configurations:

Insert arrangements shall be in accordance with figure 1.

Shell configuration shall be in accordance with figure 2.

Termination and layout arrangement shall be in accordance with figures 3 and 4

Mounting configurations shall be in accordance with figure 5.

Materials and finishes:

Mounting brackets half hard brass, UNS-C26000 in accordance with ASTM B36/B36M.

Finish in accordance with MIL-DTL-24308.

Spacer with push fastener, phosphor bronze, UNS-C54400 in accordance with ASTM B139/B139M.

Finish: Gold in accordance with MIL-DTL-45204, type II, grade C, class 1 or equivalent over copper in accordance with SAE AMS 2418.

Push fastener with mounting brackets, beryllium copper, UNS-C17300 in accordance with ASTM B196/B196M.

Finish: Gold in accordance with MIL-DTL-45204, type II, grade C, class 1 or equivalent over copper in accordance with SAE AMS 2418.

Electrical:

Size 20 signal contacts:

Current rating:	7.5 amps nominal
Initial contact resistance:	.008 ohms maximum
Proof voltage:	1000 V r.m.s.

Power contacts:

.078 dia:	10 amps
.125 dia:	40 amps
Initial contact resistance:	.0005 ohms maximum

Shielded contacts:

Nominal impedance:50 ohms

Insertion loss:	-0.46 dB at 1 GHz
	-1.5 dB at 2 GHz
VSWR:	1.15 average at 1 GHz
	1.56 average at 2 GHz
Proof voltage:	1000 V r.m.s
Initial contact resistance:	.008 ohms maximum

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TABLE I. PIN and dash numbers.

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 7)
M24308/33-001	5W5	0.078	Straight	A
M24308/33-002		0.078	Straight	D
M24308/33-003		0.078	Right angle	B
M24308/33-004		0.078	Right angle	E
M24308/33-005		0.078	Right angle	C
M24308/33-006		0.078	Right angle	F
M24308/33-007		0.125	Straight	A
M24308/33-008		0.125	Straight	D
M24308/33-009		0.125	Right angle	B
M24308/33-010		0.125	Right angle	E
M24308/33-011		0.125	Right angle	C
M24308/33-012		0.125	Right angle	F
M24308/33-013		Shielded	Straight	A
M24308/33-014		Shielded	Straight	D
M24308/33-015		Shielded	Right angle	B
M24308/33-016		Shielded	Right angle	E

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TABLE I. PIN and dash numbers - Continued.

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 5)
M24308/33-101	9W4	0.078	Straight	A
M24308/33-102		0.078	Straight	D
M24308/33-103		0.078	Right angle	B
M24308/33-104		0.078	Right angle	E
M24308/33-105		0.078	Right angle	C
M24308/33-106		0.078	Right angle	F
M24308/33-107		0.125	Straight	A
M24308/33-108		0.125	Straight	D
M24308/33-109		0.125	Right angle	B
M24308/33-110		0.125	Right angle	E
M24308/33-111		0.125	Right angle	C
M24308/33-112		0.125	Right angle	F
M24308/33-113		Shielded	Straight	A
M24308/33-114		Shielded	Straight	D
M24308/33-115		Shielded	Right angle	B
M24308/33-116		Shielded	Right angle	E

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TABLE I. PIN and dash numbers - Continued.

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 5)
M24308/33-201	13w3	0.078	Straight	A
M24308/33-202		0.078	Straight	D
M24308/33-203		0.078	Right angle	B
M24308/33-204		0.078	Right angle	E
M24308/33-205		0.078	Right angle	C
M24308/33-206		0.078	Right angle	F
M24308/33-207		0.125	Straight	A
M24308/33-208		0.125	Straight	D
M24308/33-209		0.125	Right angle	B
M24308/33-210		0.125	Right angle	E
M24308/33-211		0.125	Right angle	C
M24308/33-212		0.125	Right angle	F
M24308/33-213		Shielded	Straight	A
M24308/33-214		Shielded	Straight	D
M24308/33-215		Shielded	Right angle	B
M24308/33-216		Shielded	Right angle	E

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TABLE I. PIN and dash numbers - Continued.

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 5)
M24308/33-301	17W2	0.078	Straight	A
M24308/33-302		0.078	Straight	D
M24308/33-303		0.078	Right angle	B
M24308/33-304		0.078	Right angle	E
M24308/33-305		0.078	Right angle	C
M24308/33-306		0.078	Right angle	F
M24308/33-307		0.125	Straight	A
M24308/33-308		0.125	Straight	D
M24308/33-309		0.125	Right angle	B
M24308/33-310		0.125	Right angle	E
M24308/33-311		0.125	Right angle	C
M24308/33-312		0.125	Right angle	F
M24308/33-313		Shielded	Straight	A
M24308/33-314		Shielded	Straight	D
M24308/33-315		Shielded	Right angle	B
M24308/33-316		Shielded	Right angle	E

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TABLE I. PIN and dash numbers - Continued.

PIN	Insert arrangement (see figure 1)	Size 8 contact	Termination type	Mounting configuration (see figure 5)
M24308/33-401	21W1	0.078	Straight	A
M24308/33-402		0.078	Straight	D
M24308/33-403		0.078	Right angle	B
M24308/33-404		0.078	Right angle	E
M24308/33-405		0.078	Right angle	C
M24308/33-406		0.078	Right angle	F
M24308/33-407		0.125	Straight	A
M24308/33-408		0.125	Straight	D
M24308/33-409		0.125	Right angle	B
M24308/33-410		0.125	Right angle	E
M24308/33-411		0.125	Right angle	C
M24308/33-412		0.125	Right angle	F
M24308/33-413		Shielded	Straight	A
M24308/33-414		Shielded	Straight	D
M24308/33-415		Shielded	Right angle	B
M24308/33-416		Shielded	Right angle	E

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PIN example: M24308/33 -105 F G

Specification sheet number:

Dash number from table I:

Finish:

A = Pure electrodeposited aluminum. 1/
F = Cadmium. 1/, 5/
K = Zinc nickel. 1/ 2/
P = Passivated stainless steel. 1/
T = Nickel Fluorocarbon Polymer. 1/ 2/
Z = zinc. 1/
- = nickel 3/ or gold 4/

Class:

G, N, D or M (See MIL-DTL-24308)

NOTES:
1/ Finish for class G or N.
2/ Not for use with class N or M
3/ Finish for class D.
4/ Finish for class M.
5/ Cadmium finish may have a chromate conversion coating or chromic acid rinse which includes hexavalent chromium and are not recommended for Army ground systems.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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Referenced documents. In addition to MIL-DTL-24308, this document references the following:

MIL-DTL-45204
ASTM B36/B36M
ASTM B139/B139M
ASTM B196/B196M
SAE AMS 2418

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force – 85
NASA - NA
DLA - CC

Preparing activity:
DLA - CC

(Project: 5935-2019-131)

Review activities:
Army - AT, CR4, MI
Navy - AS, CG, MC, SH
MISC - MDA

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/ .

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